VoiceOps 2026: a voice support agent for ZentroMart

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Language model: Gemini 3.1 Flash Lite | Speech: Gnani STT and Gnani TTS (timbre-v2.5) | Search: BM25 plus character n-grams

1. What I built

I built the agent in the four steps the brief describes, each one on top of the last.

The first version is just a system prompt. It tells the model it is ZentroMart's support assistant, that it has to say it is an AI in the first reply, and that it should answer in the customer's language. It has no documents, so everything it says comes from what the model already knows.

For step 2 I cut the 21 policy documents into one chunk per section ID (like `REF-2`), keeping each document's status and date. To find chunks I mix two kinds of keyword search and then favour the more authoritative document type when two disagree, following the order in KB-00.1 (current policy, then agent guide, help, FAQ, marketing). A small word list and a query rewrite help with how customers really talk, such as "send it back", Hindi and Hinglish. The model is told to answer only from what it was given, to say so when the documents don't cover something, and to list the section IDs it used. Date sums go through a small calendar tool, because the model is bad at counting business days.

Step 3 adds memory and tools. The agent remembers the order, the problem and what has been verified, so a follow-up like "how long will that take?" works. When someone gives an order ID it calls `lookup_order`, asks for the last four phone digits or the pincode, allows one retry and then stops. `create_ticket` is used for handovers. I put the important safety rules in the code, not only in the prompt: the checking of the caller, hiding the digits they give, and never saying a ticket exists unless the tool really returned one.

Step 4 wraps this in Gnani speech. Audio goes through Gnani STT, into the same agent, and the reply comes back through Gnani TTS. Replies are short, dates include the year, amounts are spoken as rupees and order numbers are read in groups of four. If the audio is empty or unclear the agent politely asks again. `python run.py <folder>` runs a whole folder of audio files and saves the reply text, the reply audio and the chunks used, and there is a browser page with a microphone for demos.

2. Results on the practice set

I ran the 52 practice questions through the prompt-only agent and through the document-grounded one. A question counts as correct only if the decision is right, every required fact appears and nothing forbidden is said.

52 practice questions	Step 1: prompt only	Step 2: with documents
Correct answers	**12 / 52 (23%)**	**39 / 52 (75%)**
Right sections found	not applicable	100%

Topic	Step 1	Step 2	Language	Step 1	Step 2
Refund timelines	1 / 14	12 / 14	English	8 / 33	28 / 33
Return windows	3 / 11	8 / 11	Hinglish	1 / 11	6 / 11
Warranty	0 / 5	3 / 5	Hindi	3 / 8	5 / 8
Return eligibility	1 / 4	4 / 4			
Damaged items	0 / 3	1 / 3			
Not in the documents	3 / 5	2 / 5			

For the multi-turn part I ran the 23 practice conversations (96 customer turns). 83 turns were correct (86%) and 11 of the 23 conversations were right from start to finish. The expected tool call was made in 94 of 96 turns, handovers to a human were right in 4 of 6 cases, and the right sections were found 96% of the time. The weakest scenarios were return eligibility (7 of 10), compensation demands (6 of 8) and cancelling a shipped order (6 of 8).

I also checked search alone, with no language model at all. For all 47 answerable questions the right sections were in the top 5 results, and it stayed at 98% when I used the speech-style text instead of the typed question.

```
(.venv) C:\Users\Nedstark\Desktop\Capstone\voiceops-agent\voiceops-agent>python run.py chat
ZentroMart support (type 'quit' to exit, '/sources' to see the last chunks)

you> Hi, how long does a UPI refund take?
bot> I'm ZentroMart's AI support assistant. Once initiated, a UPI refund takes 1 to 3 business days to reach your accoun
t. If you're asking about a specific order, feel free to share your order ID!

you> Can I return a swimsuit that doesn't fit?
bot> Swimwear cannot be returned or exchanged for fit or sizing issues. However, if the item arrived damaged, defective,
 or wrong, it would be eligible for a full refund or replacement under our damaged items policy.

you> Can I get a student discount?
bot> I'm sorry, I don't have information about student discounts. I can connect you with a human colleague if you would
like.

you> quit
```

Terminal chat: an answer from the documents, a refusal for something that cannot be returned, and an honest "I don't have information" with an offer of a human colleague.

3. What went wrong, and what I learned

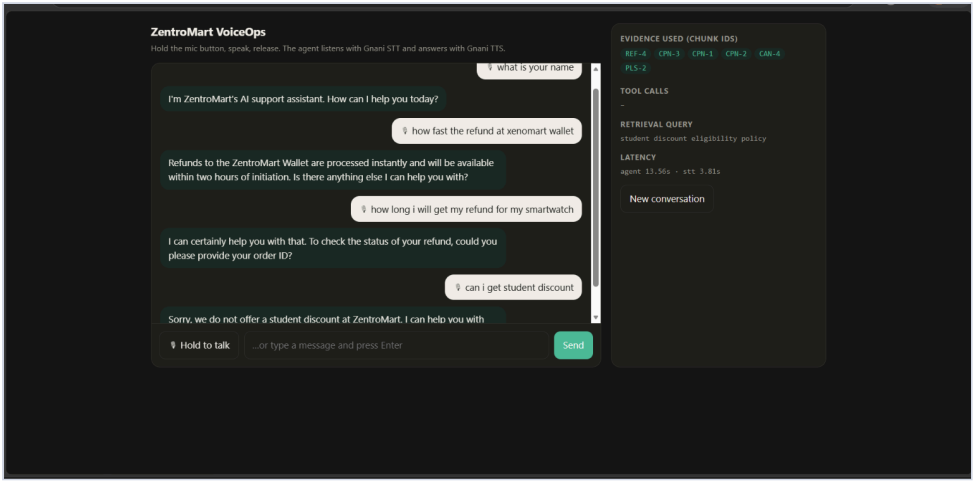
The prompt-only agent sounds sure of itself and gets the numbers wrong. It answered 1 of 14 refund-timeline questions correctly. Once it had the documents it got 12 of 14, mostly because it stopped repeating the FAQ's "5 to 7 days" for every refund and followed the policy instead.

The result I did not expect is on the five questions the documents cannot answer. The prompt-only agent got 3 and the grounded one got only 2. My demo shows the same thing: I asked the voice page whether I could get a student discount and it answered "Sorry, we do not offer a student discount at ZentroMart". No document says that. It should have said it had no information and offered a colleague, as it did when I asked in the terminal. I have not read every failed reply, so my guess is that the retrieved text makes the model feel it knows the answer, but I need to check this.

Hinglish was the weakest language (6 of 11, against 28 of 33 in English). Speech recognition also makes mistakes: in the demo it heard "xenomart" for ZentroMart. The agent still understood and answered correctly. Most failed conversation turns (11 of 13) were turn 4, the first real request after the caller was verified, and I haven't yet found the common cause.

Several problems were in my setup rather than the agent, and each took time to find. The free Gemini plan allows only 20 requests a day for each model, so my first evaluation scored 0 of 3 until I read the error text and saw it was a quota limit. One of the newer models spends part of its output limit on hidden thinking, which cut a spoken reply off in the middle of a sentence, so I gave it extra room. Gemini 3 also refuses a tool call unless a "thought signature" is sent back, so I changed the code to return it. Finally, the audio files from Gnani TTS have an unfinished length in their header, which crashed my code before speech-to-text, and I fixed that too. Each fix has a test, and all 24 tests pass without any API keys.

The agent is also slow for a phone call. In my demo the agent took about 5 to 14 seconds per reply and Gnani STT about 3 to 4 seconds.



Browser demo with a microphone. Left: the conversation (my spoken questions as heard by Gnani STT, and the replies). Right: the evidence panel with the section IDs used, tool calls, the search query and the time taken.

4. What I would do next

First I would read the failed replies and fix the "I don't know" cases, for example by saying "I don't have that information" whenever the best chunks only loosely match. Then I would add embeddings that understand Hindi and Hinglish next to the keyword search, and add more Hinglish examples. I would read every failed conversation turn to find the shared cause. I have also not yet run the whole evaluation with the speech-style text, or with real recordings from different voices and phones, so that comes next. For latency, streaming the speech in and out would help a lot.

5. Extra work

Besides the four required steps I added a browser demo with an evidence panel, recognition of SKUs that are spoken aloud ("ele zero zero one two"), speech recognition in English and Hindi, a different Gnani voice for each language, a batch runner that can resume after a stop, helper scripts that check the setup and compare models, and 24 automated tests.